

YOUSEF AHMED

AI / ML Engineer

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate (GPA 3.62) with hands-on experience building and deploying ML/DL models across computer vision and data engineering workflows. Published researcher who deployed real-time AI on edge hardware (NVIDIA Jetson Nano) achieving 96% accuracy at 20 FPS. Proficient in Python, PyTorch, TensorFlow, LangChain, and Docker, with applied knowledge of MLOps and cloud tools.

EXPERIENCE

Machine Learning Engineer • PioPetro • Ohio, USA (Remote)

Jun 2024 – Aug 2024

- Completed applied ML training on petroleum industry use cases including Exploration, Artificial Lift Technology, Well Intervention, and Well Stimulation; built predictive models for domain-specific datasets.
- Gained practical exposure to feature engineering on regression-based production forecasting.

AI/ML Intern • Information Technology Institute (ITI) • Menoufia, Egypt

Jul 2023 – Sep 2023

- Built and benchmarked 6+ supervised and unsupervised ML models (classification, regression, clustering) using Scikit-learn across structured datasets; achieved up to 93% F1 on held-out test sets.
- Applied cross-validation, hyperparameter tuning, and evaluation metrics (Accuracy, F1, RMSE) to optimize model performance.

AI/ML Intern • Digital HUB (D-HUB) • Cairo, Egypt

Aug 2023

- Built a breast cancer detection classifier achieving 95% accuracy using CNNs and Scikit-learn; awarded 3rd place in a competitive AI challenge co-hosted with the Electronics Research Institute.

PROJECTS

Agri-Bot — AI-Powered Crop Disease Detection Robot (Graduation Project)

Nov 2023 – Jun 2024

- Designed a real-time CV pipeline using YOLOv8 + MobileNetV2 on NVIDIA Jetson Nano: 96% accuracy at 20 FPS, fuzzy PID motor control, web interface for remote monitoring. Published at IUGRC 2024.
- **GitHub:** github.com/Yousef-koura/agri-bot | **Stack:** Python, PyTorch, YOLOv8, MobileNetV2, Jetson Nano

Personal RAG Chatbot

May 2026

- AI chatbot using LangChain, ChromaDB, and Groq (Llama 4 Scout), with a FastAPI backend and a custom HTML chat UI, enabling natural language Q&A over a structured personal knowledge base.
- **GitHub:** github.com/Yousef-koura/personal-rag-assistant | **Stack:** Python, LangChain, LangSmith, ChromaDB, Groq API, FastAPI

FMCG Data Engineering Pipeline

April 2026

- Built an end-to-end Lakehouse pipeline on Databricks using PySpark, Delta Lake, and Unity Catalog with Medallion Architecture (Bronze → Silver → Gold), Star Schema modeling, incremental loading, and a live Genie AI dashboard.
- **GitHub:** github.com/Yousef-koura/FMCG_Data_Engineer_Project | **Stack:** Databricks, PySpark, Delta Lake, Unity Catalog, SQL

Chest X-Ray Pneumonia Classification

May 2023

- Compared a custom CNN vs. ResNet-18 transfer learning on 5,863 chest X-rays; tackled class imbalance via pos_weight and WeightedRandomSampler — best model achieved 85.1% accuracy, 89.3% F1, and 100% pneumonia recall.
- **GitHub:** github.com/Yousef-koura/Deep_Learning_Projects/tree/main/PRJ-004_xray_CNN | **Stack:** Python, PyTorch, ResNet-18

Steganography Image Detection

Feb 2023

- Trained a CNN-based classifier to detect hidden data embedded in digital images, achieving 90% classification accuracy.
- **GitHub:** github.com/Yousef-koura/Steganography-Detector | **Stack:** Python, MobileNetV2, Flask

EDUCATION

B.Sc. Mechatronics Systems Engineering • GPA: 3.62 / 4.0

Sep 2019 – Jul 2024

MSA University (supervised by University of Greenwich) • Giza, Egypt

Military Service — Completed

Sep 2024 – Mar 2026

SKILLS

Programming: Python | C++ | SQL

ML / DL: Supervised & Unsupervised Learning | CNNs | RNNs | Transfer Learning | Object Detection (YOLO) | LLMs | RAG

Data Engineering: ETL/ELT | Medallion Architecture | Delta Lake | PySpark | Databricks | Star Schema

Frameworks: TensorFlow | PyTorch | Scikit-learn | Pandas | NumPy | FastAPI | OpenCV | LangChain | LangSmith | LangGraph

MLOps: Git | GitHub | Docker | ONNX Runtime | AWS (fundamentals)

Soft Skills: Problem-solving | Time management | Adaptability | Research

PUBLICATIONS & CERTIFICATIONS

Agri-Bot: AI-Powered Crop Disease Detection and Medication Recommendation Robot

Published at the 8th International Undergraduate Research Conference (IUGRC 8) • 2024

Machine Learning Specialization — Coursera (2023) | Python 101 for Data Science — IBM Cognitive Class (2022)